

Virtual Reality as an Automated Stress-Reduction Therapy Tool - Case Study on War Refugees

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Figure 1: The VR-based therapeutic tool combines several therapeutic methods in a virtual environment, cutting the user off from reality. The tool was evaluated in a controlled experiment among office workers and during a pilot study among Ukrainian refugees.

ABSTRACT

This case study presents research and reflections concerning Virtual Reality (VR) as an automated stress-reduction therapy tool. The study coincided with the outbreak of war in the Ukraine, thus the software was tailored to reduce the acute stress of refugees. We created a relaxation training program in a virtual, pleasant environment in the form of a comfortable mountain apartment. The pilot programme was conducted on the group of 55 Ukrainian refugees participating in up to five therapeutic sessions. The system effectiveness was evaluated by a set of sensors collecting objective physiological measures such as GSR (galvanic skin resistance) and EEG (electroencephalography). Before and after each session, the volunteers filled in questionnaires regarding their current stress level and mood, and that subjective data was juxtaposed with objective data from the sensors. In this work, we present our experience, insights and hope to inspire researchers and practitioners to explore the opportunities given by VR-supported therapy.

*Both authors contributed equally to this research.

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KEYWORDS

stress reduction, virtual reality, affective computing, bilateral stimulation, refugees, relaxation techniques

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1 INTRODUCTION

The term ‘stress,’ initially introduced to scientific literature in the 1930s, found its way into everyday language in the 1970s [14]. Presently, it is commonly employed to describe various unpleasant emotional states such as frustration, anger, dilemmas, feeling overwhelmed, or exhaustion. Despite its widespread use, the precise definition of ‘stress’ remains challenging and somewhat ambiguous [8]. According to WHO, stress is defined as any type of change that causes physical, emotional, or psychological strain [19]. Stress is the human body’s response to an interference that requires attention or action [20]. Individuals undergo both favorable and adverse emotional states, which exert an influence on their interpersonal dynamics, perception of the world, and overall state of well-being. The manner in which an individual reacts to stressful situations plays a pivotal role in determining their quality of life and overall health. Stress triggers changes in vital bodily functions. When facing danger, the sympathetic system becomes active leading to increased adrenaline secretion, elevated blood pressure, faster heart

rate, higher blood glucose levels, and reduced digestion [16]. Stress primarily impacts the nervous system, especially the hypothalamus and pituitary gland, prompting a "fight or flight" response. Short-term stress readies the body for action, but prolonged stress can be harmful. Acute stress symptoms typically fade quickly, depending on a person's ability to cope. Untreated stress can worsen, straining the nervous system and causing physical problems [3], emphasizing the importance of stress therapy and treatment. Stressors, which are events disrupting our physical or social equilibrium, form an integral part of the stress definition. They can stem from physical sources like overcrowding, disease, or natural disasters, as well as social sources like unemployment, family issues, loss of a loved one, or crime. Stressors also vary in intensity. The mildest ones are everyday nuisances, often problematic due to their frequency (e.g., minor conflicts or lateness). The next level comprises severe stressors tied to significant life changes, such as marriage, job transitions, or the addition of a family member. The highest level includes catastrophic stressors, affecting entire communities, like wars, natural disasters, or terrorist attacks, which threaten fundamental human needs and values, such as life and safety. The stress accompanying these events is usually severe, sometimes traumatic, and can have enduring psychological effects [15], as is the case of war refugees.

The conflict in Ukraine commenced on February 24, 2022, leading to a significant influx of refugees into neighboring countries, primarily women and children (the government in Kyiv prohibited men of draft age from leaving Ukraine). As a result, over 3 million refugees arrived in Poland in a brief span [13]. Many women and children faced an entirely unfamiliar environment, often lacking basic personal items, money, or a secure place to stay. Separated from their loved ones who remained in war-torn cities, these individuals experienced high levels of stress.

Leveraging prior experience with VR for stress reduction [9], we created a Ukrainian-language VR therapy app. This app incorporates relaxation techniques like breathing exercises, visualization, and bilateral stimulation (BLS). To enhance user experience, we integrated minimally invasive biomedical sensors (dry frontal EEG, GSR, and eye movement tracers) for continuous monitoring. User interaction is natural, relying on gestures and biofeedback. Participants completed pre and post-session questionnaires on their mental health, stress levels, and mood. This case study shares our long-term research experience with a VR therapeutic tool, offering insights, reflections, and highlighting potential study management risks. The rest of the paper is organized as follows. Section 2 presents the related studies investigating the effect of VR on stress reduction. Section 3 presents the VR therapeutic tool, the course of the experiment, and its results. Section 4 discusses the insights and reflections resulting from the project and provides some final remarks.

2 BACKGROUND

Most studies investigating the effect of VR on stress reduction focus on immersing the user in a new virtual environment, without any additional tasks or stimuli. For example, in [10] during the session, the users found themselves strolling through a trekking course with natural landscape and a soothing background soundtrack. The authors of the study juxtaposed data on subjective stress

reduction with physiological parameters when VR or biofeedback treatment was applied. 74 volunteers took part in the experiment in the Clinical Trial Center in Samsung Medical Center. Head mounted display (HMD) utilized in this study was Samsung Gear VR, sensors attached to the subject's body collected biofeedback (alternative medicine approaches that enhance the ability to influence body systems based on indications from measuring devices eg. EEG) and heart rate variability (HRV) parameters data, and several different psychological tests were used as a reference. The concluding observation was that although both VR relaxation and biofeedback significantly facilitated stress reduction among the users, the difference between the treatments was not substantial.

Another study that shared similar characteristics was conducted in [4] and aimed to assess the efficacy of VR technology in stress reduction among night-shift anesthesiologists. A group of 30 professionals was randomly divided into an intervention group, which would be subjects to VR immersion and a control group. The results indicate that VR experience can reduce the stress and tiredness stemming from the long working hours and allows the subjects to rest. In [1], green meadows VR experience was provided for NHS (the National Health Service) clinicians during their workday. There, both subjective and objective measures were taken in the form of emotional state assessment, physiological arousal, and heart rate. The study was conducted on 39 clinicians and obtained data indicates that VR immersion increases the subject's mood by increasing happiness and relaxation, at the same time reducing anger, sadness and anxiety. What is more, there was a visible correlation between subjective assessment and heart rate reduction.

There is no doubt that the COVID-19 pandemic has had a negative effect on society on the whole and increased stress levels globally, especially among essential and front-line healthcare workers. In [2] a 3-minute VR simulation of a nature scene was applied to decrease subjective stress levels among COVID-19 treatment units employees in the United States. 100 participants equipped with Oculus Go or Pico G2 4K HMDs took a stroll in a 360-degree simulation of a soothing and tranquil landscape. The subjects assessed their stress level before the simulation using a 10-point scale. The results clearly showed the reduction of subjective stress level after the VR experience, however the researchers themselves are aware that more research has to be done to fully prove the efficacy of the solution, including both subjective and objective stress measures, and control conditions.

In [18], the researchers compared the effects of VR relaxation with traditional relaxation exercises among patients with psychiatric disorders ranging from anxiety to psychotic, depressive or bipolar disorders. One again, the VR simulation comprised 360-degree natural scenery video. However, in this scenario some interactive elements were added such as guided meditation and progressive muscle relaxation. The trial was conducted on 50 patients who were randomly divided into an intervention group that relaxed using VR and a control group who followed a standard relaxation routine for 10 days. Perceived stress was measured before and after the session based on eight visual analog scales of momentary negative and positive affective states. The results were very promising, demonstrating that VR-based therapy might be an effective, user-friendly, self-managed relaxation solution to reinforce the efficiency of psychiatric treatment.

The area where the research on VR-assisted mental health treatment is still lacking is the examination of changes in affective (emotional) reactions of VR users. Most of the studies utilize questionnaires, which are analyzed in detail [17], still remaining merely a subjective assessment. There are but a few works that use objective measurements obtained from sensors. For example, Dongrae et al. [5] analyzed a photoplethysmogram (optical measurement of changes in organ volume), electrodermal activity (the variation of the electrical properties of the skin in response to sweat secretion), heart-rate variability, skin conductance and temperature of twelve healthy participants performing mental tasks using VR to monitor stress. A kernel-based extreme-learning machine (type of neural networks) was used to classify five different levels of stress situations: baseline, mild stress, moderate stress, severe stress and recovery. The average classification accuracy utilizing K-ELM and extracted features was approximately 95%. In [6] the authors propose a Decision Support System (DSS) to automatically classify stress levels during a VR experience. The system collates ECG, RIP and EEG sensors data as well as body gestures data tracked with Kinect tool. A ground truth was established by using self-rated and clinical-rated stress levels. During the session, detected stress level (ranging from 0 to 1) is reported to the therapist and graphically displayed on a diagram, which can be accessed via web interface. It must be remarked that in the latter case virtual reality experience was delivered via a monitor screen, not a head mounted display.

The examples provided above indicate that although the development of new virtual environments is rapid, the clinical use of VR is still virtually non-existent. What the technology needs to be more readily adopted in the clinical environment is reliable, long-term research proving its sustainable effect on chronic stress, instead of one-off experiments. Secondly, VR therapeutic scenarios should evolve from passive experiences of calming environments into interactive immersions. The third issue is connected with establishing real efficacy of used solutions, which means relying not only on subjective evaluation of the participant. Thus, we still feel there is a knowledge gap in the discussed topic, especially concerning the above-mentioned issues.

3 VR THERAPEUTIC TOOL

3.1 VR application

As we mentioned in the Introduction, this is not our first approach to VR-based stress reduction. Our first application, focused on stress reduction among office workers, was in the form of a walk in the woods, where the user performed exercises at successive stops. Although the stress level was reduced among most participants, 7 participants reacted negatively to the session or fell asleep during the experience. Therefore, we decided to conduct an extensive interview with those participants and consult them during the development of the new release of our application. We intended to make improvements that wouldn't significantly affect the methodology used in the application but would allow for more promising results for different users. The most significant changes (determined during the interviews) introduced concerning the previous version of the application are presented in Tab. 1.

The proposed version of the VR application is in the form of a cozy apartment in the mountains (see Fig. 2). During the session,

participants relax on a comfortable chair (the same is mapped in the VR environment) in any convenient position (see Fig. 3a). The main scene is calibrated automatically (based on the user's head position) - the main element (a window with the mountain view) of the scene is centered vertically and horizontally. The narrator assists throughout the session, acting as a therapist.

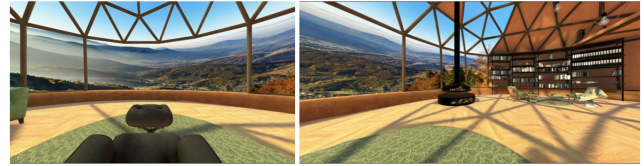


Figure 2: VR environment in the form of a cozy apartment in the mountains. The real chair was mapped in the VR environment identical.

A short adaptation scene precedes the therapy scene. First, the user is asked to look around the apartment and get comfortable in the chair. Then, the user moves seamlessly into the therapy sessions, which involve various relaxation techniques such as breathing methods, visualization, and bilateral stimulation.

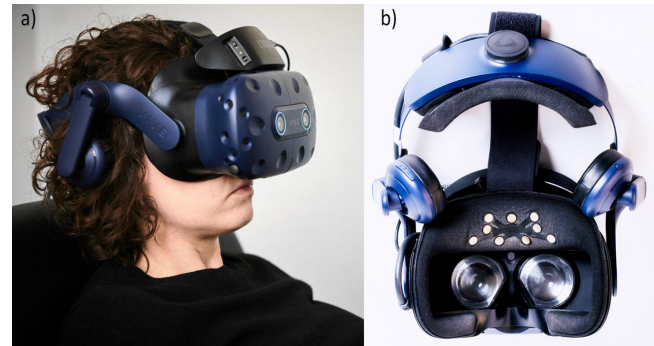


Figure 3: a) A snapshot of a subject during a VR therapeutic session. b) Looxid Link - VR compatible EEG sensor allowing to receive biometric signals in real time.

3.2 Stress monitoring methods

The sessions were monitored using EEG and GSR sensors. For this specific group, we eliminated the cumbersome EEG wet cap (gel or saline liquid is used to increase the conductivity value) by replacing it with a discreet, HDM-integrated, dry EEG Looxid Link (see Fig. 3) at the expense of reducing the number of channels - 6 channels with sensors mounted on the user's frontal lobe. The EEG system remains almost unnoticeable. Users did not complain about the inconvenience (preparation for the session, soaking the hair, and the need to wash the hair afterward). We used also a GSR sensor with convenient mounting on the user's fingers. Those sensors allow the recording of raw measurement data and real-time monitoring of interpretation of the user's involvement and stress level. Before and after the session, users fill out General Health Status Questionnaires (GHQ12 [7] and GMS [12]) so that we know

Table 1: The most significant changes introduced concerning the previous version of the application. Issue - comments made/observed during the previous experiment (among office workers); Solution - changes made to a new version of the software; Result - the impact of the solution on the experiment (among war refugees).

Issue	Implemented solution
Falling asleep during the session	Solution: We introduced eye tracking to monitor the correctness of performed exercises. If the exercise is performed incorrectly or the eyes are closed, the "awakening" procedure is triggered (visual and voice messages), and a less intense version of the scenario is performed. Result: Cases of falling asleep have been virtually eliminated.
Distractions, boredom	Solution: We changed the surroundings: less detailed, reduced image artifacts (unwanted distortion / anomaly - caused by the animated complex natural landscape, grass, and trees) introduced a dominant tertiary environment of the mountain landscape, and the environment was reduced to a minimum while maintaining excellent resolution and realism. The role and guidance of the speaker leading the session has been expanded. Result: Users reported no distraction issues related to the graphical environment rendering, and no one complained about imperfections in rendering environmental elements.
Inconvenience associated with wet EEG cap	Solution: We eliminated the cumbersome EEG wet cap by replacing it with a discreet, HDM integrated, dry EEG system (at the expense of reducing the number of channels - 6 channels with sensors mounted on the user's frontal lobe). Result: The EEG system remains almost unnoticeable. Users have stopped complaining about the inconvenience (preparation for the session, soaking the hair, and the need to wash the hair after the session).
Discomfort associated with the use of multifunctional chest and EMG sensors	Solution: The use of cumbersome sensors mounted directly on the user's body has been avoided. Only the GSR sensor with convenient mounting on the user's fingers was left. Result: Sensors (GSR) remain almost unnoticeable.
Automatic movement in virtual forest vs. no movement in reality	Solution: Virtual walking through the forest scenery has been replaced by a static view of the mountain panorama and the cozy surroundings of a comfortable apartment. The user sits on a comfortable chair during the whole session. They have the ability to rotate their heads freely and look around the apartment in a natural way but in a sedentary position. Result: VR experience according to users is very realistic and predictable, no one reported problems with confusion and lack of understanding in the narrative and flow of the session.

their subjective feelings about their mental state at that moment, the intensity of stress and the emotions accompanying the moment. The system's unique feature is the capacity to compare data recorded during the user's subsequent therapy sessions. Thus, the therapist can easily monitor the progress and achieve effects. These tools in the pilot study phase facilitated us in objectively evaluating the effectiveness of the application.

3.3 Course of the experiment

At the end of 2021, we began work on improving the system based on the measurement experience we gathered and user interviews. Our interventions were evolutionary: each implemented improvement was tested by a group of volunteers who took part in our previous research. Once the implementation was verified and accepted, we implemented the next one. The final version coincided in time with the start of the war on Ukrainian territory. Local media (newspapers, online portals, radio and television) played a vital dissemination role.

Both the media and our observations of refugees arriving in Poland inspired us to adapt the developed application for such users. We initiated extensive cooperation with local city authorities as well as non-governmental organizations involved in assisting refugees from Ukraine. We modified the content of the narratives - the narrator's voice was lent by a professional native Ukrainian speaker. When the application was ready for use in March 2022, we

launched an information campaign together with Lodz city council. To increase the level of trust, we hired a Ukrainian refugee who was in charge of recruiting subjects for VR sessions. With limited equipment resources (one complete station) and limited availability of the premises, during three months, we managed to organize more than 250 sessions, which lasted from 45 to 60 minutes each.

The study protocol was approved by the local Ethical Committee (resolution no. 3/2021 dated October 21, 2021). Before each session, users were informed about the course of the VR experience and the exclusionary conditions, and provided informed consent. The entire course of the session was monitored by the supervisory staff. We assumed that each participant would be able to benefit from five therapeutic sessions (one session per week). Unfortunately, due to personal reasons and relocation, not everyone has completed the session sequence.

Finally, we recruited $n = 55$ participants, 52 female, 3 male, aged $M = 35$, $SD = 10.93$. The study was conducted in a period when the university was closed (there were no students), in a quiet, air-conditioned room. We provided a shopping voucher for the equivalent of USD 20 as remuneration for participating in one session.

3.4 Results

To verify the effectiveness of the presented application, we used pre and post-tests (subjective measurement) and the GSR and EEG signal (objective measurement). The results obtained from pre and

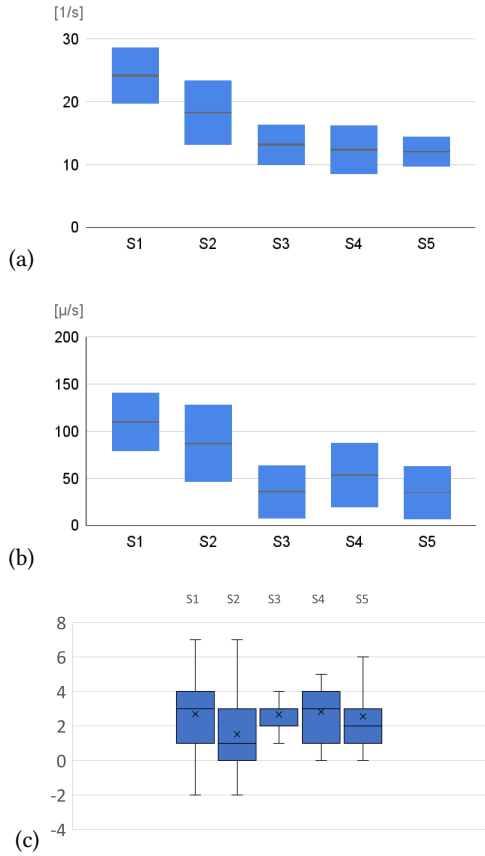


Figure 4: The box plot of a group analysis (average across all subjects for each session) of (a) resistance gain over time and of (b) relaxation level (obtained from EEG) gain over time. (c) Changes in the declared level of stress based on pre-and post-session surveys.

post-test, in the form of mean stress level, are presented in Fig. 4. First, we smoothed GSR signal with a moving average (30 samples) to reduce measurement noise. Next, to determine the change in signal waveforms at the beginning and end of the session, the group analysis (average across all subjects for each session) in the form of mean and standard deviation values of resistance gain over time is presented in Fig. 4a. Since the Looxid Link provides analytical information such as relaxation level (in a range of 0 to 1) over time, similarly as in the case of GSR, first we smoothed those information with a moving average (30 samples). Then, we determined the change at the beginning and end of the session (average across all subjects for each session) in the form of mean and standard deviation values of relaxation level gain over time. The results are presented in Fig. 4b

4 REFLECTIONS

Having completed the pilot study, we are able to offer our insights and reflections resulting from the project. These insights come from our knowledge of system design, its creation, and development, our

observations during testing, implementation, and 250 sessions with the final users (55 refugees from Ukraine). Our insights are framed by four overarching themes: the recruitment process, the course of the study, study results, and consequence associated with the project.

4.1 Reflections on the recruitment process

In order to recruit refugees, we faced a significant challenge in effectively engaging potential participants. When we began the recruitment process, we initially assumed it would be relatively easy, thinking that extending a helping hand and offering support to address apparent mental health issues using proven therapeutic methods in a more user-friendly VR environment would be well received. However, this assumption turned out to be far from reality. We underestimated the deep distrust and skepticism of refugees who found themselves in a foreign country with limited resources and often incomplete family units. Despite the warm initial welcome by Polish society, Ukrainian refugee communities were initially closed, suspicious, and paralyzed by fear of meeting their basic existential needs. In the early stages of their stay, refugees frequently changed their place of residence in order to find one that would guarantee the satisfaction of their basic needs. Other priorities and concerns overshadowed therapeutic considerations. Various channels were used to reach our target group, including city authorities, official government refugee registration offices, non-governmental organizations providing refugee assistance, local media and social media platforms. Direct access to refugee communities proved challenging. These groups were tight-knit and suspicious of strangers, often using communication channels that were inaccessible to us. Surprisingly, we were bombarded with numerous questions, suggesting that even the subject of therapeutic assistance at a national university (sessions were organized at the Lodz University of Technology) had to be sufficiently credible. Potential therapy participants had doubts about whether it was "human experimentation", they knew nothing about VR technology and considered it a dubious machine that could harm them. Overcoming these barriers to recruiting early participants became a formidable challenge. One of our team members invited refugees into their home to build trust. As a result, they agreed to attend the first sessions. The idea of formally hiring one of them to help with the project emerged and became a turning point. Someone with a background in war refugee communities found it much easier to connect with the target groups, largely due to direct access through social media and the common language, eliminating language and cultural barriers. The city authorities played a crucial role in organizing an information campaign using local media and providing us with municipal office space for two weeks, which greatly enhanced the credibility of the project. From that point on, the recruitment process became much more effective. The hired Ukrainian effectively handled new enrollments by fully explaining the project. Still, some uncertainty remained among the target group consisting mainly of women. They preferred to attend the VR sessions in pairs or bigger groups to have the sense of security and companionship. This experience highlighted the importance of a rational approach to the existing hierarchy of needs within the target group when planning outreach programs. It was also a reminder that providing the subjects with a sense of security

and safety should be a priority. Another conclusion is that finding effective ways to reach the desired community directly is essential to the success of any relief program.

4.2 Reflections on the course of the study

Some of the refugees, for reasons beyond our control, did not complete the full series of tests. Only 30 subjects participated in all offered sessions. 6 relocated and were not able to continue participation due to a very inconvenient commute. 8 people returned to their country of origin. 11 people, without giving a reason, did not report back to participate in further sessions. Based on behavioral observation and analyzing data from VR therapy sessions, we noticed that virtually 100% of the subjects participating in the first study showed signs of heightened distrust, skepticism and critical attitude towards our initiative. These perceptions changed after subsequent sessions - with each visit we could observe much more engagement and willingness to participate in the study. In comparison to Polish office workers who did not suffer any trauma, we recognized the refugees as more willing to continue the sessions because they clearly provided them with comfort and much needed stress reduction. Only two of the recruited subjects had any previous exposure to a VR experience. Participants of the first sessions in the conducted interview made it clear that they were astonished by the level of immersion and very keen on exploring the presented VR environment. It is likely that for this reason, the expected measurable objective therapeutic effect was obtained only during the second and subsequent sessions (89% of cases).

The recording of biomedical signals did not cause any critical problems. Thanks to the adopted procedure of placing the GSR sensors on the index and ring finger of the left hand, we did not have any case of shortened contacts or discontinuity of measured quantity. Similarly, we did not encounter any issues with recording eye tracking signals and HMD localization (6 degrees of freedom - the object can move forward/backward, up/down, and left/right; it can also rotate around three different axes: yaw, pitch, and roll). In contrast, monitoring EEG signals proved to be more troublesome. Since the vast majority of participants were women, applied face makeup was a source of some problems, i.e. dry Looxid electrodes sometimes lost good connection with the skin. In one case, getting a stable EEG signal was impossible due to the very "low" forehead of the participant. It is worth mentioning that the HMD mounting procedure included the need to expose the forehead by moving away the fringe as contact of hair with EEG electrodes virtually always leads to temporary or permanent loss of signal.

We would also like to discuss some organizational issues that we faced. First of all, we were initially too optimistic about scheduling therapy sessions. In spite of our best efforts, it was challenging to keep the sessions on track. Being in an unfamiliar city, suffering from communication problems, and having no experience in using local public transport caused trouble when reaching our University, and thus significant delays in our schedule. Since we often had to plan the session cycle in the evenings, there was a rather unfortunate situation of accidentally setting off the Institute's monitoring system, triggering a loud acoustic alarm and, more concerning for

participants, activating the intervention procedure for an "unauthorized intruder in the building." This experience was particularly unpleasant for individuals with a history of war refugee trauma.

What is more, the rise of national threat level connected with the situation in Ukraine, caused increased security measures to be introduced in public institutions. The participants had to go through security and identify themselves, which often caused additional setbacks. We have noticed a lack of empathy from the facility's security personnel. We mention these incidents to underscore the diverse situations we may encounter in practice. Following each occurrence, we implemented interventions to adjust our organizational procedures, ensuring avoiding similar issues in the future. This approach proved effective in resolving and mitigating the following challenges.

Another unforeseen logistic problem was that women often attended the sessions accompanied by their young children. Providing the children with care and entertainment during their guardian's session was an additional challenge. Fortunately, we swiftly organized a set of building blocks and provided colorful crayons and markers.

4.3 Reflections on the study results

For the vast majority of measurements, skin resistance increased steadily during each session. The intensity of this increase decreased with each successive session. This may indicate that the initial stress level for subsequent sessions is lower. In addition, we noticed that the trends among all participants were more consistent with each session. Similarly, measurements from the Looxid device indicate a state of meditation or drowsiness among participants, which was especially noticeable in the period of BLS (which shares many features with a meditative state). This positive condition deepens from one session to another session.

The results were promising and showed that VR-based BLS is efficient in mood improvement and stress reduction. These effects are also related to a decrease in the subjective perception of stress levels. Data analyses showed that the repetition of VR training does positively affect its efficacy in symptom reduction which is relatively stable from one session to another. Taking into account the special sample we worked with, our results are more than promising. However we need to mention some limitations of our pilot study. The first group of limitations concerns the sample. Firstly, it was a sample of random volunteers who responded to media announcements. Secondly, participants were under the constant pressure of uncontrollable events and sudden demands and possibilities which were impossible to control in the study but they might affect our results. The third issue is the fact that the group was relatively small, with a high dropout rate. Another group of limitations concerns the way we measured mental health status and study design. Although GHQ-12 is a well recognized and reliable measure, possessing an independent clinical diagnosis of mental health would be an additional advantage. Moreover, taking into account the complexity of uncontrollable problems, challenges, stress and trauma faced by our participants - case-control design of study might increase the reliability of our result. However, despite the limitations mentioned above, the results are very promising and guide

our imagination to the era of affordable, accessible, home-based treatment of adverse mental health symptoms.

We did not find any negative effects of VR stress management sessions in terms of VR sickness symptoms. Experimentally we asked the participants to fill in VRSQ [11] before each session to check if they experienced any oculomotor or disorientation symptoms. It appeared that the number of negative symptoms of VR sickness reported before the session is significantly higher than that reported after experienced immersion.

4.4 Reflections on the consequences associated with the project

Thanks to the press conference and various media publications in newspapers, the Internet, radio and television, we had no problems recruiting volunteers for our study. Additionally, we were contacted by many NGOs interested in implementing the application at their spot. Since our solution was designed for research, it was rather unsuitable for such applications: the hardware is expensive and additional trained personnel is required. Thus, we prepared a simplified version dedicated to the Oculus Quest 2 HMD, supplemented with a self-service IU. This version has been designed for individual users or patients receiving therapy sessions at a therapy clinic, at home, or refugee assistance center. Since standard Oculus Quest HMD is not equipped with additional EEG, eye tracking, and GSR sensors, this version supports only one selected scenario and the amount of recorded data has been naturally limited. We believe that the therapist will be able to conduct a control session and adjust home therapy. Despite significant functional limitations, the application still fulfills its primary mission of stress reduction.

Despite the great initial interest in the software, ultimately only two foundations decided to implement it at their sites: Masz Wszystko w Sobie and Together-Razem Centre. The first one, with the help of donors, purchased 5 Oculus Quest sets and have been working with since then. The other one wishes to use the expanded version of the application and is waiting for financial support. The most obvious parties interested in the application - refugee aid centers - unfortunately did not have facilities or personnel to make use of the software. Despite initial intentions, when conversation reached the issue of financial and human resources, the organizations began to withdraw their interest. We have also received information about the need to implement the system in elementary and secondary schools due to the fact that many children have problems with adaptation. Unfortunately, after consultations with a child psychologist, it turned out that the created scenario will not fulfill its purpose. We are in the process of preparing an application tailored for children in school age, which will be focused on stress reduction, but in a more dynamic, task-oriented environment.

After reading one of article about our project, the board of a significant VR goggles distribution company in the US decided to provide us with VR kits in any quantity to work with refugees. Unfortunately, due to procedural reasons, the implementation of this initiative faced considerable delays. Local legal and tax regulations further posed challenges to this endeavors. Thus, this valuable initiative eventually met an unfortunate end.

5 CONCLUSION

The drama of the war on Ukraine and the massive refugee inflow prompted many Poles to help the displaced. As a scientific team with considerable experience in the field of stress reduction using immersive technologies, we decided to adapt our previous solution to those specific conditions. Several meetings with charitable organizations' representatives directly involved in helping refugees allowed us to develop a therapeutic scenario to reduce severe post-traumatic stress. Promotion organized by the City of Lodz Office and local media contributed to the project's indisputable success. The complete therapeutic sequence was successfully carried out on 30 participants within three months. The therapeutic VR tool we produced is still being used by some Polish NGOs. And we, the creators, have gained a new portion of precious experience and research data. The message of our case study is: developed solutions can be and should be continuously adapted to the circumstances.

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